Medicaid Drug Utilization Analysis

CS 133 Project Report

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Abstract

This project presents comparative analysis of prescription drug utilization and Medicaid reimbursement patterns across the United States using the 2025 State Drug Utilization Data (SDUD) published by the centers of Medicare and Medicaid services (CMS). The dataset comprises approximately 2.6 million records of 2025, covering drug reimbursement submissions from state Medicaid agencies nationwide.

Our analysis addresses 5 core questions: Identifying the most commonly prescribed drugs for Medicaid patients across states, quantifying government reimbursements to pharmacies on a quarterly basis, ranking the top 10 drugs by both volume and total reimbursement value, exploring how pharmaceutical supply chains might be informed by prescription trends, and examining the proportion of drug categories prescribed across different states.

Key findings reveal that BIKTARVV, an HIV antiretroviral, leads all drugs in total Medicaid reimbursement at approximately $3.1 billion despite relatively modest prescription counts, reflecting its extremely high per-unit cost. GLP-1 class drugs including Ozemmpic, Zepbound, Trulcity and Jardiance collectively account for 17.2% of total Medicaid reimbursements across the 21 states analysed, with California recording the highest GLP-1 spending of any state. Analysis of prescription volume proportion across states confirms that larger states such as California and New York drive a disproportionately large share of national Medicaid prescription volume.A small number of high cost specialty drugs drive a disproportionately large share of total Medicaid expenditure, a pattern that holds consistently across states.

To support predictive analysis, we trained three machine learning classifiers to predict whether a GLP-1 drug record falls into a high or low Medicaid coverage bucket, defined as above or below 80% Medicaid coverage of total reimbursement. All three models used class_weight = 'balanced' to address the inherent 90%/10% class imbalance in the data set. The Random Forest classifier achieved the strongest overall performance with 95.5% accuracy, a ROC-AUC of 0.99 and a macro F1 of 0.89, while Logistic Regression achieved 86.0% accuracy and macro F1 of 0.76. Across all three models, unit reimbursement, number of prescriptions, and state-level indicators were the most influential predictors, suggesting that drug utilization volume and geographic context are the primary drivers of Medicaid coverage patterns of GLP-1 medications.

Background & Problem Definition

This project uses the Centers for Medicare & Medicaid Services (CMS) State Drug Utilization Data (SDUD) 2025 dataset, which is a publicly available resource that logs prescription drug utilization and reimbursement information across Medicaid programs in the US. The dataset contains about 2.6 million rows and 15 columns which comes out to about 250 MB of data. It includes both quantitative and categorical data. The data is structured in a CSV format. This dataset was selected because it provides a comprehensive coverage of Medicaid activity.

The overall problem explored in this project is understanding how prescription drug usage and spending vary across the US, with an extra focus on GLP-1 drugs. Prescription drugs account for a significant portion of healthcare expenditures, and identifying patterns in utilization and reimbursement can help inform cost management, policy, and supply issues as well.

The five main questions we wanted to look at were:

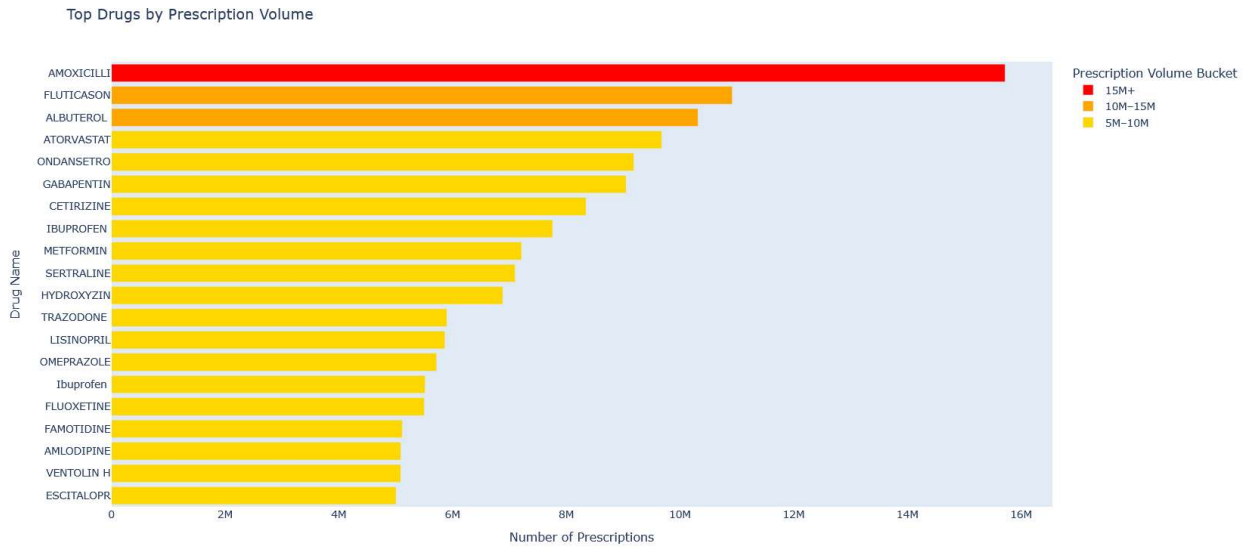
1. What are the top 10 most prescribed drugs across all states?
2. What is the proportion of drug prescriptions across different states?
3. Which drugs cost the government the most money?
4. How are GLP-1 drugs being prescribed across different states?
5. Which states have the highest Medicaid utilization of GLP-1 drugs?

Exploratory data analysis and visualizations play a big role in answering these questions and preparing the data for machine learning. The visualizations created during this project help identify trends in prescription growth, cost, and geographic variation. These insights are essential for identifying meaningful features that can also later be used in predictive models.

This problem is relevant because Medicaid serves millions of Americans across the country. Understanding patterns can support better allocation of resources, help drive medical innovation, and contribute to policy decisions as well, especially in growing categories like GLP-1 treatments.

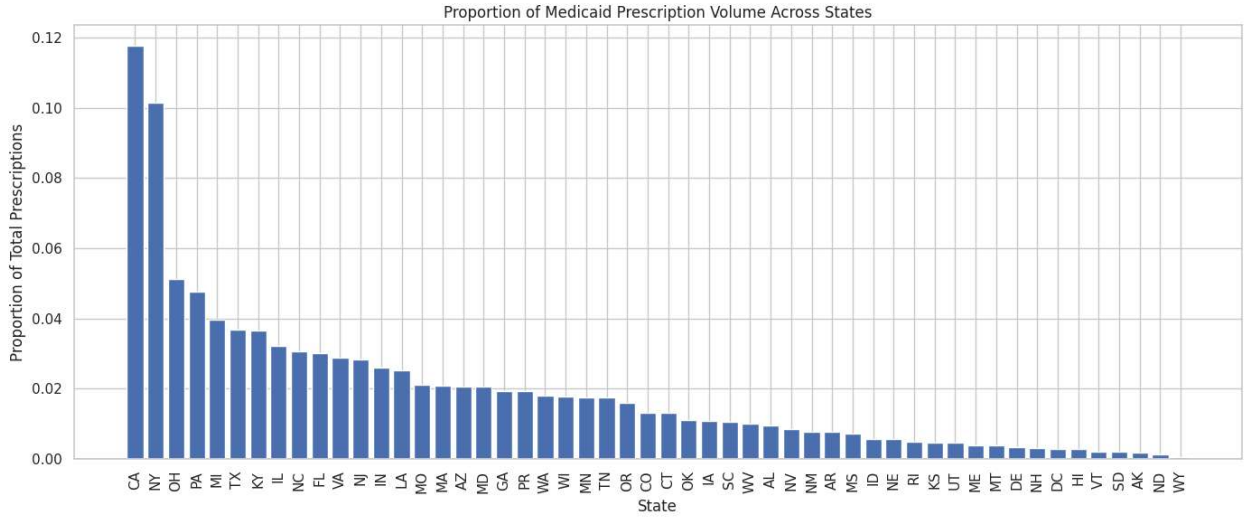
Data Visualization & Interactivity

What are the top 10 most prescribed drugs across all states?



This graph ranks drugs by volume. Each bar represents a drug while the length of the bar equals the total amount of drugs prescribed. The graph is sorted from highest to lowest. The graph also is colored into buckets with the buckets being 5-10 million, 10-15 million, and 15 million and more. This is then color coded and a legend is included in the top right to increase interpretability. The graph is interactive, and if it is run and visualized locally provides more information. When the mouse hovers over a bar in the graph, it shows the name of the drug, the total drugs prescribed, and the bucket it belongs to.

What is the proportion of drug prescriptions across different states?

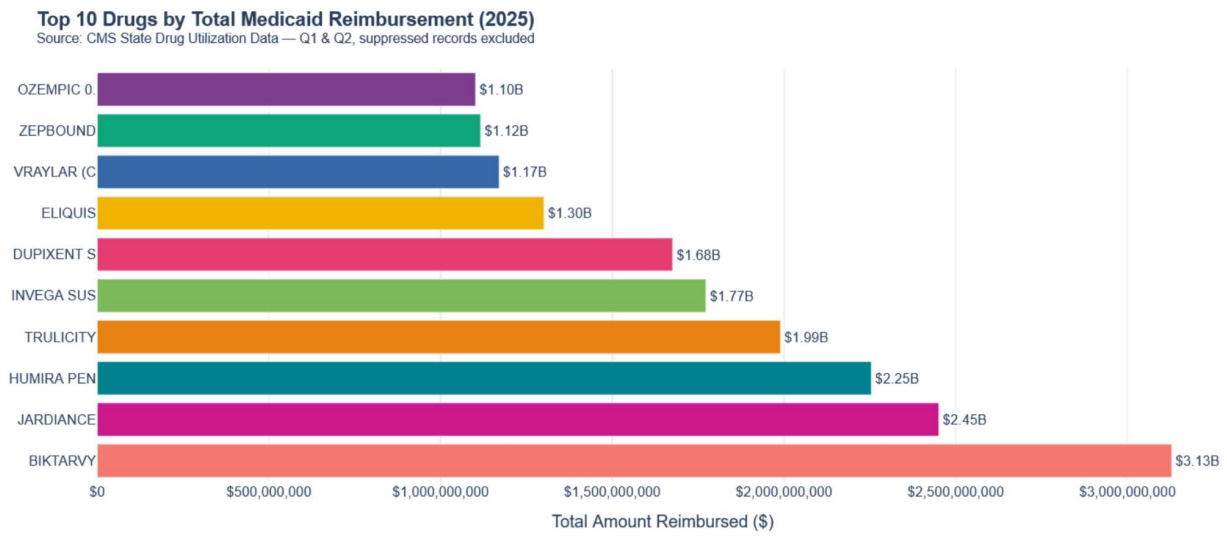


The graph shows the following equation graphed.

Proportion of Total Prescriptions = (Prescriptions in that state)/(Total prescriptions across all states)

From the analysis, California has a much higher proportion of the total share of Medicaid usage.

Which drugs cost the government the most money? (Interactive)

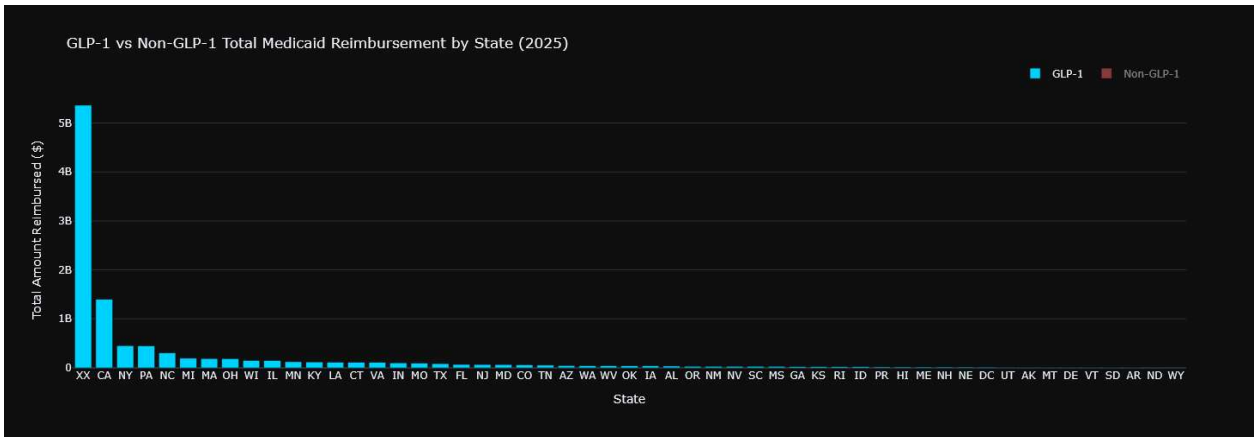


The interactive chart above ranks the top 10 drugs by total Medicaid reimbursement amount across all states combined. Only non-Suppressed records are included, leaving approximately 1.3 million clean records for this analysis.

The chart included two tabs. The overview tab shows all 10 drugs sorted by total reimbursement, with each drug assigned a distinct color. Clicking any bar switched to the State Breakdown tab, which shows how reimbursement for that drug is distributed across individual states. A dropdown allows filtering by a single state, and a sort toggle switches the ranking between total reimbursement amount and prescription count for that drug or state.

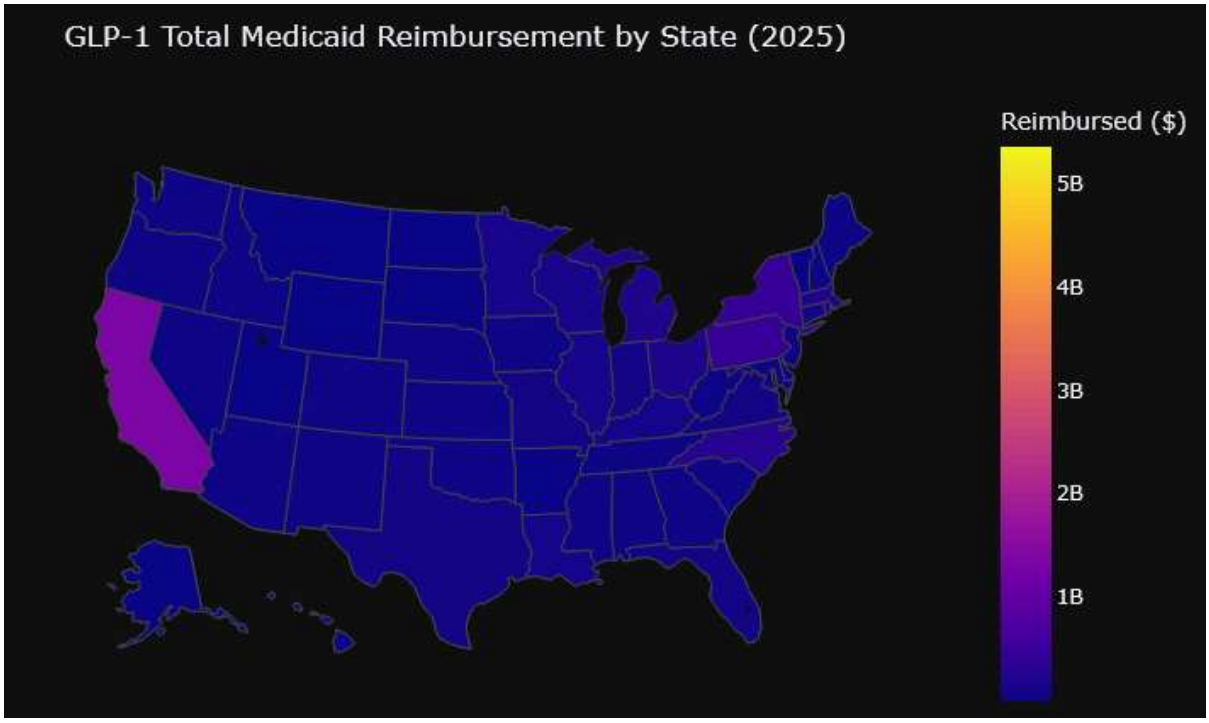
BIKTARVY, an HIV antiretroviral, leads all drugs and approximately $3.1 billion in total Medicaid reimbursement despite having relatively few prescriptions compared to other top drugs. This reflects its extremely high cost per unit. JARDIANCE(empagliflozin, a diabetes/heart failure medication) ranks second at $2.5 billion with over 2.8 million prescriptions, making it the highest volume drug in the top 10. GLP-1 related drugs Ozempic, Zepbound, and Trulicity all appear in the top 10 confirming the significance and growing financial footprint of this drug class within Medicaid. Notably, when sorted by prescription count rather than reimbursement, the ranking shifts considerably –HARDIANCE and ELIQUIS move up while BIKTARVY drops, illustrating that high spending does not always correspond to high prescription volume.

How are GLP-1 drugs being prescribed across different states?



The graph above shows how CA has the most usage of GLP-1 drugs. The graph is sorted from highest to lowest. The graph is interactive, and if it is run and visualized locally provides more information. When the mouse hovers over a bar in the graph, it shows the name of the drug, the total drugs prescribed, the total cost in reimbursement for Medicaid, and the total share of GLP-1 drugs compared to total drugs.

Which states have the highest Medicaid utilization of GLP-1 drugs? (Interactive)



The highest reimbursement rate for GLP-1 drugs is in CA followed by MA with a considerable drop. Even though GLP-1 reimbursement is the highest in CA, it is still about 5 billions dollars less than non GLP-1 Medicaid reimbursements.

Machine Learning Pipeline & Evaluation

Data Preparation & Pipeline

The dataset we decided to use initially has about 2.6 million rows of data covering Medicaid reimbursement on prescription drugs across every state in the US. However, we wanted to focus on GLP-1 drugs only to learn its reimbursement patterns and uncover its insights.

To start off, we create a list of commonly prescribed GLP-1 drugs manually and filter the rows that specifically contain listed GLP-1 drugs. This resulted in about approximately 2000 rows of GLP-1 related data, among which approximately 27.9% were suppressed due to HIPAA privacy regulations. The final GLP-1 dataset contains 1455 rows.

Next, we standardized the Product Name column by writing a function that strips dosage information from drug names. For example, entries like 'MOUNJARO 1', 'MOUNJARO 5', and 'MOUNJARO 7' were all consolidated into a single 'MOUNJARO' entry. This ensures the model treats all dosage variants of the same drug as one unified feature rather than separate drugs.

To prepare the data for machine learning, we decided to engineer a target variable called Coverage Bucket. We used the Total Amount Reimbursed and Medicaid Amount Reimbursed to calculate the coverage rate.

Coverage Rate = Medicaid Amount Reimbursed / Total Amount Reimbursed

This ratio represents the proportion of the amount that is covered by Medicaid. Then, we converted this continuous ratio in a binary classification target.

1 as High: Medicaid covers more than 80% of the reimbursement
2 as Low/Medium: Medicaid covers 80% or less of the reimbursement

To prevent data leakage, the original reimbursement amount columns (Total Amount Reimbursed, Medicaid Amount Reimbursed, Non Medicaid Amount Reimbursed) were dropped after computing the target variable. The final dataset of 1455 rows was split into 80% training set and 20 % testing set. We also applied stratification to ensure the class distribution of the target variable has preserved the same high and low distribution.

Preprocessing was handled using sklearn's ColumnTransformer, which applied two transformations in parallel. StandardScaler was applied to two numerical features (Units Reimbursed and Number of Prescriptions) to normalize their scale. We also applied OneHotEncoder to the four categorical features(State, Product, Name, Utilization Type and Quarter) so the model can process them.

Model Training

We decided to use Logistic Regression, Decision Tree, and Random Forest to predict Medicaid GLP-1 coverage classification. All three models were trained on the same training dataset containing 1164 rows and 37 encoded features. To address the class imbalance, we also decided to apply class_weight='balanced' to penalize more on the minority classes.

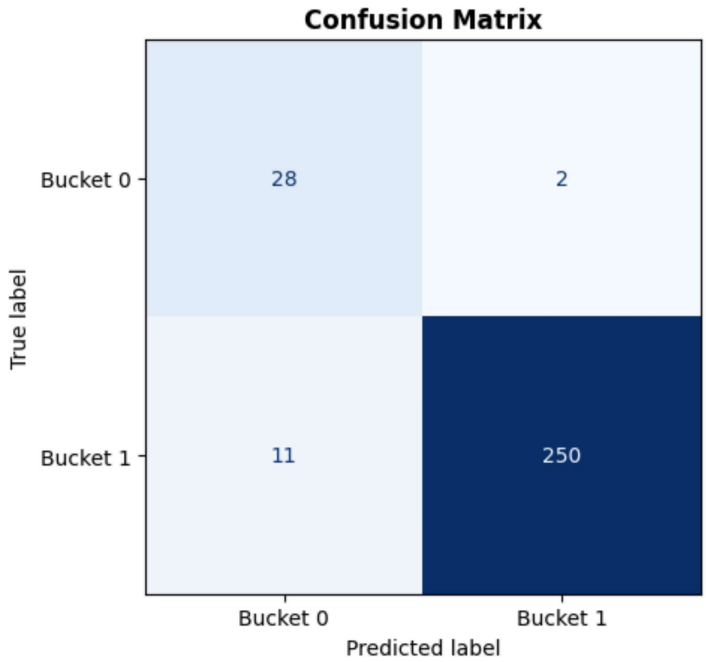
We also performed 5-fold cross validation using Macro F1 as the scoring metric. We chose Macro F1 score as it treats both classes equally regardless of their size, thereby ensuring a fair and honest metric. The model with the highest mean Macro F1 score across the 5 folds was selected as the best performing model and subsequently evaluated on the test set for final performance.

Evaluation results:

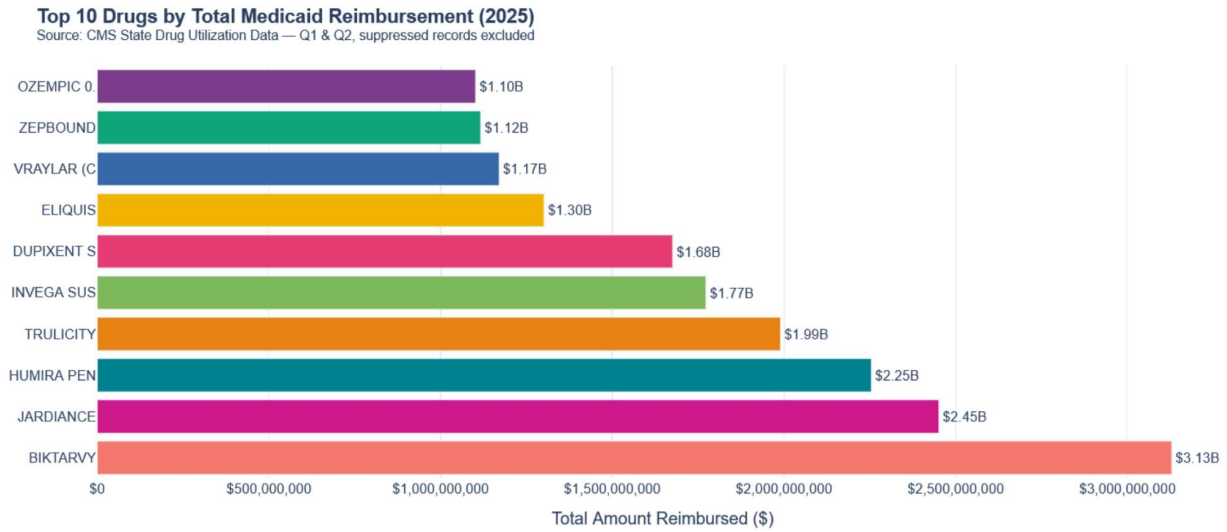
Model	Macro F1	Accuracy	F1 High	F1 Low/Medium	ROC-AUC
Logistic Regression	0.780	86%	0.92	0.60	0.9645
Random Forest	0.8931	95%	0.9747	0.8116	0.991
Decision Tree	0.7920	90%	0.9421	0.6420	0.895

Model Evaluation

We decided to proceed with Random Forest since it outperformed both models across every metric. Below, we have the confusion matrix of Random Forest performed on our test set. The model achieved a ROC-AUC score of 0.991, which indicates near perfect separation between the two classes. It's also correctly identified 28 out of 30 Low/Medium coverage cases with a recall score of 0.93. The model learnt that States matter more than drugs. This signals the importance of where GLP-1 prescription is filled matters more than the GLP-1 drug itself when it comes to Medicaid coverage. This also suggests that some states cover more of the GLP-1 drug costs through Medicaid than others. We can conclude that this reflects real life where different states have their own set of rules for what Medicaid will cover or not.


Confusion Matrix
Bucket 0
Bucket 1
True label
28
2
11
250
Bucket 0
Bucket 1
Predicted label

Results & Code Interpretation



This chart shows the top drugs by reimbursement amount with Medicaid across the country. This highlights how a significant concentration of spending is among a small number of drugs with the highest-ranking drugs accounting for a large portion of the costs. This indicated that the spending is driven by a few high cost medications rather than being evenly distributed.

```
df =
pd.read_csv("https://download.medicaid.gov/data/sdud-2025-updated-dec2025.c
sv")

# Remove suppressed rows
df = df[df["Suppression Used"] == False].copy()

# Normalize fields
df["Drug"] = df["Product Name"].str.upper().str.strip()
df["State"] = df["State"].str.strip()
```

This code block from the TopDSpending.ipnyb file in GitHub demonstrates the cleaning process we used. It was first cleaned by removing suppressed records and standardizing drug and state names. This ensures that aggregation is accurate and avoids any duplicate entries that might be caused by formatting errors.

```
top10 = (
```

```
df.groupby("Drug")
.agg(
Total_Reimbursed=("Total Amount Reimbursed", "sum"),
Total_Rx=("Number of Prescriptions", "sum"),
Total_Units=("Units Reimbursed", "sum"),
)
.nlargest(10, "Total_Reimbursed")
.reset_index()
)
```

The data was grouped by drug name and aggregated to compute total reimbursement, number of prescriptions, and total units reimbursed. The top 10 drugs were then selected by total reimbursement which allowed identification on which medications contribute the most to spending.

```
fig_overview = px.bar(
    top10.sort_values("Total_Reimbursed"),
    x="Total_Reimbursed",
    y="Drug",
    orientation="h",
    color="Drug",
    text=top10["Total_Reimbursed"],
    title="Top 10 Drugs by Total Medicaid Reimbursement (2025)",
)

fig_overview.show()
```

A horizontal bar chart was used to visualize the ranking. This format made it easy to compare the spending across drugs.

For the model we decided to use the parameter class_weight = 'balanced' to address class imbalance in the dataset. Since 90% of the data belonged to the 'high' class, a model could achieve high accuracy by predicting 'high' for every case. By applying the balanced class weights, the model gives more importance to minority classes. This leads to a more reliable and fair model. Below is the code snippet showing this.

```
full_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', LogisticRegression(class_weight='balanced',
max_iter=1000, random_state=42))
```

```
])
```

We believe that the Random Forest model deserves an emphasis in this section since it outperformed the two other models in all evaluation metrics. From the classification report below, we can observe that it achieved 96% overall accuracy. The minority class also received a recall of 0.93, meaning it correctly identified 28 out of 30 low.

```
              precision    recall  f1-score   support

    Bucket 0       0.72      0.93      0.81        30
    Bucket 1       0.99      0.96      0.97       261

    accuracy                           0.96       291
   macro avg       0.86      0.95      0.89       291
weighted avg       0.96      0.96      0.96       291

ROC-AUC Score: 0.9906
```

References

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